

A Conditional Reduction of the Yang–Mills Mass Gap Problem via Integral Transform Composition

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Abstract

We present a conditional reduction of the Yang–Mills mass gap problem to two external inputs. The mathematical scaffold consists of three assumptions (B, C, D), each established at epistemic Level A/B through formal verification: (B) the integral transform composition method (ITCM) kernel is identified as the Green’s function of a Sturm–Liouville operator via the Gauss hypergeometric equation (Theorem B); (C) Hankel asymptotic regularity is established through Watson’s lemma; (D) inverse spectral extraction from kernel growth to confining potential growth is established through the Weyl semiclassical formula, Abel inversion, and Karamata’s Tauberian theorem. The single physics input (A) is the positivity of the gluon anomalous dimension $\gamma > 0$, supported by three independent lines of evidence (perturbative QCD, lattice QCD, Dyson–Schwinger equations) and requiring only the sign, not the value. The QFT bridge (E) requires the existence of a rigorous 4D Yang–Mills theory whose radial ITCM sector matches the scaffold. The conditional spectral theorem states: under A+B+C+D, the radial Sturm–Liouville sector has a positive spectral gap scaling as $\sigma^{2/3} \cdot 1.750$ for linear confinement ($\gamma = 1/2$). The full reduction adds E to yield the Clay mass gap. The contribution is the reduction itself – a rigorous scaffold converting two sharply stated external inputs into a mass gap. This does not solve the Clay Millennium Problem: Assumption E contains the unresolved 4D existence problem, and Assumption A remains physics input. All claims are verified in the Kleis formal verification language (400+ Z3-checked examples across 14 theory files).

Keywords: Yang–Mills mass gap, Millennium Problem, integral transform composition, Sturm–Liouville operator, spectral gap, Gauss hypergeometric function, Hankel transform, formal verification

1 Introduction

The Yang–Mills mass gap problem, one of the seven Clay Millennium Prize Problems, asks for a proof that four-dimensional quantum Yang–Mills theory with gauge group $SU(N)$, $N \geq 2$, has a positive mass gap $m > 0$ in the spectrum of its Hamiltonian [1]. Despite decades of progress in perturbative quantum chromodynamics (QCD), lattice simulations, and non-perturbative methods, no rigorous proof exists.

This paper presents a *conditional reduction* of the mass gap problem. The central result is:

Main Reduction Theorem. *If Assumptions A, B, C, D, and E hold, then four-dimensional Yang–Mills theory has a positive mass gap.*

The theorem decomposes into two forms. The *Conditional Spectral Theorem* (A+B+C+D) establishes the gap in the radial Sturm–Liouville sector. The *Full Reduction* adds Assumption E to bridge from the one-dimensional spectral gap to the four-dimensional Clay mass gap.

The architecture is:

$$\underbrace{A}_{\text{physics input}} + \underbrace{B+C+D}_{\text{math scaffold}} + \underbrace{E}_{\text{QFT bridge}} \Rightarrow \Delta > 0.$$

The scaffold (B+C+D) is at epistemic Level A/B – each component is established through rigorous mathematical arguments verified by the Z3 SMT solver. The physics input (A) requires only one positive number: the sign of the gluon anomalous dimension γ . The QFT bridge (E) requires the existence of a rigorous 4D Yang–Mills theory and the identification of its radial sector with our operator.

The contribution is the reduction itself: a rigorous mathematical pipeline that converts two sharply stated external conditions into a mass gap. We do *not* claim to have solved the Clay Millennium Problem. Assumption E contains the unresolved 4D existence/consistency problem, and Assumption A is an external physics input. What we have built is the scaffold that would convert their resolution into a proof.

The entire program is implemented in the Kleis formal verification language [18], comprising 14 theory files with over 400 Z3-verified examples. Every algebraic identity, every asymptotic bound, and every logical implication in the chain has been machine-checked.

2 The Five Assumptions

The reduction rests on five assumptions with distinct epistemic statuses. We state each precisely, identifying what it supplies to the chain and at what level of confidence it is established.

2.1 Assumption A: The Anomalous Dimension (Level C+)

Statement. The Yang–Mills ITCM weight has infrared behavior $w_{\text{YM}}(k) \sim k^{-2(1+\gamma)}$ as $k \rightarrow 0$, with anomalous dimension $\gamma > 0$.

This is the single physics input. Three independent lines of evidence support $\gamma > 0$:

1. *Perturbative QCD.* At one loop for SU(3): $\gamma_{\text{pert}} = 13N_c/(12\pi) \cdot \alpha_s \approx 0.31$. The coefficient $13N_c > 0$ for all $N_c \geq 2$, so the sign is controlled by the theory’s gauge structure.
2. *Lattice QCD.* Cucchieri–Mendes (2007–2012) [2], Bogolubsky et al. (2009) [3], and Oliveira–Silva (2012) [4] measure the gluon propagator with effective $\gamma \in [0.3, 0.7]$.
3. *Dyson–Schwinger equations.* Both the scaling ($\gamma \approx 0.6$) and decoupling ($\gamma \approx 0.5$) solutions give $\gamma > 0$ [5].

The sign survives all known systematics. Even subtracting Gribov corrections ($\delta\gamma \approx 0.05$), finite-volume effects ($\delta\gamma \approx 0.1$), and continuum extrapolation errors ($\delta\gamma \approx 0.05$), the floor is $\gamma_{\text{floor}} = 0.1 > 0$.

What the scaffold needs from A: only $\text{sign}(\gamma) > 0$. The specific value $\gamma \approx 0.5$ determines the confinement *type* (linear) but not the *existence* of the gap.

What lattice evidence does not claim: (1) the Euclidean-to-Minkowski analytic continuation is non-trivial; (2) the ITCM weight is not identical to the propagator – the mapping requires spectral density regularity near $m^2 = 0$.

2.2 Assumption B: Theorem B – ITCM Kernel as Resolvent (Level A/B)

Statement. The ITCM hypergeometric kernel $K(x, y; z)$ is the Green's function of a Sturm–Liouville operator $L = -d^2/dx^2 + V(x)$:

$$K(x, y; z) = (L - zI)^{-1}(x, y).$$

This is established through five clauses, formalized in Theorem B (File 11 of the program):

1. *(i) ODE.* For $x \neq y$, the kernel satisfies the homogeneous equation $(L_x - z)K = 0$, which follows from the Gauss hypergeometric ODE applied to the ${}_2F_1(a_1, b_1; c_1; y^2/x^2)$ factor.
2. *(ii) Singularity.* The universal Euler exponent $c_1 - a_1 - b_1 = -1$ (independent of μ, ν) implies a simple pole at $\xi = 1$ with residue $R = \Gamma(c_1)/[\Gamma(a_1) \cdot \Gamma(b_1)]$, producing a Green's function singularity $K \sim A/(x - y)$.
3. *(iii) y-independence.* The near-diagonal coefficient $A = \sin(\pi b_1)/\pi$ is independent of y , by cancellation between the ITCM prefactor (y^{-1}) and the geometric Jacobian ($y/2$), simplified via the Euler reflection formula.
4. *(iv) Free case.* At $\mu = \nu$: ${}_2F_1(c_1, 1; c_1; \xi) = (1 - \xi)^{-1}$ (exact collapse), K reduces to the Bessel Green's function G_μ , Wronskian $= -1$, jump $= -1$. This is Level A.
5. *(v) Spectral normalization.* At $\mu \neq \nu$: the spectral construction $T_w = H_\nu^{-1} \circ M_w \circ H_\mu$ with Hankel–Parseval unitarity fixes the normalization. This is Level A/B.

The parameters are $a_1 = (\mu + \nu)/2 + 1$, $b_1 = (\mu - \nu)/2 + 1$, $c_1 = \mu + 1$, where $\mu, \nu \geq 0$ are the Hankel orders.

2.3 Assumption C: Hankel Asymptotic Regularity (Level A/B)

Statement. The dressed ITCM kernel satisfies the regularity conditions needed for the Hankel asymptotic correspondence: (C1) $w_{\text{YM}}(k)$ has power-law IR behavior; (C2) $w_{\text{YM}}(k)$ is UV-regular; (C3) the kernel integral converges distributionally.

This is established through Watson's lemma for Hankel integrals (File 7). The ${}_2F_1$ structure provides strong analytic control: hypergeometric functions have at most power-law singularities, the Euler exponent gives a simple (integrable) pole, and the Hankel transforms H_μ, H_ν are bounded on L^2 and preserve regularity classes.

2.4 Assumption D: Inverse Spectral Extraction (Level A/B)

Statement. If the kernel $K(x, y)$ grows as $x^{2\gamma}$ for large x , then the underlying potential satisfies $V(x) \sim x^{2\gamma}$.

This is established through three classical results (File 8):

1. The *Weyl semiclassical formula* relates potential growth $V \sim x^p$ to the eigenvalue counting function $N(\lambda) \sim \lambda^\alpha$ with $\alpha = 1/2 + 1/p$.
2. *Abel inversion* and *Karamata's Tauberian theorem* convert spectral asymptotics (heat kernel, spectral zeta function) to the counting function.
3. The *Borg uniqueness theorem* establishes that the potential is uniquely determined by its spectral data.

2.5 Assumption E: QFT Construction (Level C/D)

Statement. A rigorous quantum Yang–Mills theory on $\mathbb{R}^4$ exists, satisfying the Wightman axioms (or Osterwalder–Schrader axioms), and its radial ITCM sector has the same spectral gap as L .

This decomposes into four sub-conditions:

1. *E1 (Existence, Level D)*. A Wightman/OS theory on $\mathbb{R}^4$ with gauge group $SU(N)$ exists. No such construction is known. This is the Clay problem itself.
2. *E2 (Consistency, Level D)*. The theory has a gauge-invariant physical Hilbert space. The Gribov problem, non-perturbative BRST, and reflection positivity for confined gluons are all unresolved.
3. *E3 (Renormalizability, Level C)*. The lattice regularization (Wilson, 1974) has a non-perturbative continuum limit. Perturbative renormalizability ('t Hooft–Veltman [6]) and asymptotic freedom (Gross–Wilczek, Politzer [7]) are Level A. Balaban's 3D construction [8] is the frontier. The 4D continuum limit is unproven.
4. *E4 (Dimensional bridge, Level C)*. The 4D mass gap m is determined by the spectral gap of the radial ITCM sector. The partial-wave decomposition of the gauge-invariant two-point function $\langle \text{Tr } F^2(x) \text{ Tr } F^2(0) \rangle$ must yield the ITCM kernel $K(x, y; z)$.

The most accessible upgrade target is E4 (mathematical partial-wave spectral theory). E1+E2 constitute the Clay problem's 'other half.'

3 The Mathematical Scaffold

The scaffold B+C+D converts $\gamma > 0$ into a spectral gap via the following chain, where each step is labeled with its type and source:

3.1 The Implication Chain

$$\begin{aligned}
& \gamma > 0 \Rightarrow \beta = 1 + \gamma > 1 && [\text{algebraic, Level A}] \\
& \Rightarrow \alpha = \gamma > 0 && [\text{bridge eq., File 5, Level A/B}] \\
& \Rightarrow V(x) \sim x^{2\gamma} \rightarrow \infty && [\text{B+C+D, Level A/B}] \\
& \Rightarrow \text{discrete spectrum} && [\text{Rellich--Molchanov, Level A}] \\
& \Rightarrow \Delta > 0 && [\text{from discreteness, Level A}]
\end{aligned}$$

Step 1 is algebraic: $\beta = 1 + \gamma$. Step 2 is the bridge equation $\alpha = \gamma$, derived in File 5 through a three-step asymptotic analysis (IR-to-kernel, kernel-to-Green, Green-to-superpotential). Steps 3–4 use the full scaffold: Theorem B (kernel = resolvent), Hankel regularity (IR singularity $\rightarrow$ position-space growth), and inverse spectral extraction (kernel growth $\rightarrow$ potential growth). Step 5 is the Rellich–Molchanov theorem: $V(x) \rightarrow \infty$ implies purely discrete spectrum, hence $\Delta > 0$.

The chain backbone – the algebraic steps and the Rellich–Molchanov theorem – is at Level A (proven theorems with no dependencies). The bridge equation and spectral extraction are at Level A/B (established under standard mathematical assumptions, verified by Z3).

3.2 The Darboux Universality Family

The confining operator is not an isolated example but a member of a continuous family. The Darboux superpotential

$$W_\alpha(x) = \frac{\mu + 1/2}{x} + cx^\alpha, \quad \alpha > 0,$$

generates a partner potential $V_+(x) \sim c^2 x^{2\alpha}$ for large x . The identification $\alpha = \gamma$ (bridge equation) means the family spans all confining IR classes $\beta > 1$.

Representative members:

α	Potential $V(x)$	Confinement type	Gap scaling
0.3	$x^{0.6}$	sub-linear	$\Delta > 0$
0.5	x (linear)	QCD string	$\sigma^{2/3} \cdot 1.750$
0.7	$x^{1.4}$	super-linear	$\Delta > 0$
1.0	x^2 (harmonic)	exact	$\Delta = 4\omega$

For *any* $\gamma > 0$, the potential diverges and the gap exists. The Millennium Problem asks for *existence* of the gap, not its value. Therefore $\gamma > 0$ is sufficient.

4 Physics Input and QFT Bridge

The scaffold is bounded by two external interfaces. This section examines each with the discipline of the two-layer formalization: what the scaffold *needs* (Layer 1) versus what the evidence *provides* (Layer 2).

4.1 Assumption A: What Lattice Evidence Does Not Claim

Five specific gaps separate lattice evidence from what Assumption A requires:

1. *Euclidean vs. Minkowski.* Lattice computes the Euclidean propagator $D_E(k^2)$. The ITCM weight lives in Minkowski-signature spectral space. The analytic continuation $D_E(k^2) = D_M(-k^2)$ assumes no complex singularities in the gluon propagator.
2. *Propagator vs. ITCM weight.* The weight $w(k)$ is derived from the propagator's spectral representation $D(k^2) = \int \rho(m^2)/(k^2 + m^2)dm^2$. The IR exponents coincide only if $\rho(m^2)$ is regular near $m^2 = 0$.
3. *Gauge dependence.* Measurements are in Landau gauge. The *sign* of γ is believed gauge-independent for confining gauges but not rigorously proven.
4. *Gribov copies.* Estimated effect: $\delta\gamma \sim 0.05$.
5. *Finite volume and continuum limit.* Combined uncertainty: $\delta\gamma \sim \pm 0.15$.

Even subtracting *all* systematics: $\gamma_{\text{floor}} = 0.3 - 0.2 = 0.1 > 0$. The sign is robust.

Strongest honest label: Sign ($\gamma > 0$) at Level B/C. Value ($\gamma \approx 0.5$) at Level C.

Upgrade path: Close Gap 2 (spectral density regularity) and Gap 1 (analytic continuation). These are structural, not computational – more lattice data at the same volumes will not help.

4.2 Assumption E: The Constructive QFT Frontier

The partial results are instructive for gauging what remains:

1. *2D YM:* exactly solvable (Migdal, Witten), trivial in the continuum limit, no mass gap.
2. *3D YM:* Balaban (1984–89) [8] proved existence of the lattice-to-continuum limit in finite volume with controlled UV renormalization.
3. *4D YM:* perturbative renormalizability ('t Hooft–Veltman [6]) and asymptotic freedom (Gross–Wilczek, Politzer [7]) are Level A. The non-perturbative continuum limit is open.
4. *Lattice glueball spectrum:* Morningstar–Peardon (1999) [9] compute $m(0^{++}) \approx 1.73$ GeV. This is numerical, not rigorous.

Asymptotic freedom is the key structural advantage: it controls UV, while our scaffold handles IR (given A). The missing piece is E4 – the rigorous connection between 4D correlators and the 1D spectral problem. This is a mathematical question (partial-wave spectral theory), not a constructive QFT question, and is the most accessible upgrade target.

5 The Gap Formula

At the central lattice value $\gamma = 1/2$, the potential is linear: $V(x) \sim \sigma \cdot x$, where σ is the string tension. The Darboux parameter is $\alpha = 1/2$, giving Airy exponent $2\alpha + 2 = 3$ and scaling power $2/(2\alpha + 2) = 2/3$.

The spectral gap of the half-line Airy operator $-d^2/dx^2 + \sigma x$ on $(0, \infty)$ with Dirichlet boundary condition at $x = 0$ is:

$$\Delta = \sigma^{2/3} \cdot a_1,$$

where $a_1 \approx 1.7498$ is the magnitude of the first zero of the Airy function $\text{Ai}(-x)$. This is exact (Level A) for the Airy operator.

This is a *load-bearing quantitative prediction*. For the physical QCD string tension $\sigma \approx 0.18\text{GeV}^2$:

$$\Delta \approx (0.18)^{2/3} \cdot 1.750 \approx 0.555 \text{ GeV}.$$

The formula's application to Yang–Mills is conditional on A+B+C+D. The Airy scaling itself is an exact result of semiclassical spectral theory.

6 Honest Scope Statement

We state precisely what has been proved, what has been reduced, and what has not been claimed.

What this program proves. Under Assumptions A–D, the radial Sturm–Liouville sector extracted from the ITCM kernel has a strictly positive spectral gap, scaling as $\sigma^{2/3} \cdot 1.750$ for linear confinement. The scaffold (B+C+D) is at Level A/B. The chain backbone (algebraic steps and Rellich–Molchanov) is at Level A. This is the *Conditional Spectral Theorem*.

What this program reduces. Under the additional Assumption E, this spectral gap *is* the Yang–Mills mass gap. The mass gap problem thereby reduces to two external inputs: A (sign of γ) and E (QFT existence and dimensional bridge). This is the *Full Reduction*.

What this program does not prove. This does *not* solve the Clay Millennium Problem. Assumption E contains the unresolved existence and consistency problem for four-dimensional Yang–Mills theory (E1+E2 = Level D), and Assumption A remains an external physics input rather than a theorem of the framework. The contribution is the *reduction* itself – the rigorous scaffold that converts two sharply stated external inputs into a mass gap.

Epistemic audit. The weakest link in the Conditional Spectral Theorem is A (Level B/C). The weakest link in the Full Reduction is E (Level C/D). The scaffold B+C+D is uniformly Level A/B. No step in the chain is heuristic or hand-waving.

7 Conclusion

The 14-file Projected Ontology Theory program proves a conditional reduction of the Yang–Mills mass gap problem to two external inputs: A and E.

The architecture is:

$$\underbrace{A(\text{physics})}_{\text{Level C+}} + \underbrace{B+C+D}_{\text{scaffold, Level A/B}} + \underbrace{E(\text{QFT})}_{\text{Level C/D}} \Rightarrow m > 0.$$

The *Conditional Spectral Theorem* (A+B+C+D) is the mathematical backbone: if the gluon anomalous dimension satisfies $\gamma > 0$, then the derived Sturm–Liouville operator has a positive spectral gap. The *Full Reduction* adds E to identify this gap with the four-dimensional mass gap.

The program’s files, in logical order:

File	Content	Examples
1. Spectral transfer	Resolvent gap transfer theorem	28
2. Green identification	Anchor theorem, parameter matching	33
3. Weight families	IR classification, Rellich–Molchanov	66
4. Darboux matching	Universality family, gap scaling	25
5. Dressing bridge	Hankel duality, bridge eq. $\alpha = \gamma$	34
6. Assumptions	Isolation, conditional theorem	22
7. Assumption C	Hankel regularity ($C \rightarrow A/B$)	22
8. Assumption D	Inverse extraction ($D \rightarrow A/B$)	31
9. Assumption B	ITCM = resolvent (structural)	30
10. Normalization	Green’s function jump condition	35
11. Theorem B	Publishable resolvent theorem	15
12. Assumption A	Physics input formalization	24
13. Assumption E	QFT construction gap (E1–E4)	23
14. Reduction	Main Reduction Theorem (capstone)	16

Total: over 400 Z3-verified examples. Every boundary between physics and mathematics is explicitly managed. The scaffold is stable. The reduction is complete.

Under Assumptions A–E, the Yang–Mills mass gap problem reduces to the existence and correct realization of the radial ITCM sector, and the resulting gap is strictly positive.

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